

Homework 1

MATH 220

Problem 1 (Examples of metric spaces)

Prove that the following are metric spaces

- i) $(\mathbb{R}^n, d_\infty)$ where $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$ for all $x, y \in \mathbb{R}^n$
- ii) $(C([a, b]), d_\infty)$ where $d_\infty(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$ for all $f, g \in C([a, b])$

Problem 2 (Examples of open sets in metric spaces)

- i) Let $X \neq \emptyset$ and d be the discrete metric on X . Prove that any subset of X is open.
- ii) Prove that $\{(x_1, x_2) \in \mathbb{R}^2 \mid 0 < x_1 < 1\}$ is open in $(\mathbb{R}^2, d_\infty)$ where $d_\infty(x, y) = \max_{i=1,2} |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$
- iii) Prove that $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 > 0\}$ is open in $(\mathbb{R}^2, d_1)$ where $d_1(x, y) = \sum_{i=1}^2 |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$

Problem 3 Determine whether the following sets are open and prove your claims.

- i) $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 \leq 1\}$ in $(\mathbb{R}^2, d_\infty)$ where $d_\infty(x, y) = \max_{i=1,2} |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$
- ii) $\bigcap_{n \in \mathbb{N}}]n^{-2}, 2[$ in $\mathbb{R}$ equipped with the Euclidean metric

Problem 4 (In a metric space, open balls are open indeed (btw, this is a comp problem))

Let (X, d) be a metric space, $x_0 \in X$, $r > 0$. Prove that $B_r(x_0)$ is open.

Problem 5 (Equivalent metrics)

Let $X \neq \emptyset$. We say that two metrics d_1 and d_2 on X are **equivalent** if there exist constants $c, C > 0$ such that

$$cd_2(x, y) \leq d_1(x, y) \leq Cd_2(x, y)$$

for all $x, y \in X$.

- i) Assume that d_1 and d_2 are two equivalent metrics on $X \neq \emptyset$ and let $A \subseteq X$. Prove that A is open in (X, d_2) if and only if A is open in (X, d_1) .
- ii) Let d be a metric on $X \neq \emptyset$ and for $\lambda > 0$ define the function $d_\lambda(x, y) = \lambda d(x, y)$ for all $x, y \in X$. Prove that d_λ is a metric on X that is equivalent to d .